

The Energy-Efficiency Elasticity of Durable Demand

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Abstract

Energy-efficiency adoption by households is a micro margin with macro consequences for aggregate energy demand, expenditure shares, and welfare. I quantify the energy-efficiency elasticity of durable demand in a heterogeneous-agent Bewley–Aiyagari model with lumpy replacement, fixed adjustment costs, and endogenous utilization. Two durable types (baseline vs. efficient) are priced along an empirically disciplined efficiency–price locus. Calibrated to U.S. room air-conditioner data, the model matches ownership and spending moments and delivers aggregate counterfactuals. A 10% improvement in efficiency raises adoption of efficient durables by 9% even when purchase prices rise accordingly, reduces residential energy use by 2%, and increases welfare. This framework supplies policy-relevant elasticities and sufficient statistics that can be embedded in DSGE environments to evaluate efficiency standards, rebates, and shocks to energy prices.

Keywords: energy efficiency, durable goods, heterogeneous agents, rebound effect.

JEL classifications: D15, E21, Q41, Q48, Q58.

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1 Introduction

Since 1990, the Intergovernmental Panel on Climate Change has repeatedly emphasized the need to reduce greenhouse gas (GHG) emissions (IPCC, 1990; IPCC, 1995; IPCC, 2001; IPCC, 2007; IPCC, 2014). A central lever is the adoption of energy-efficient durables that deliver the same services at lower energy intensity. Because energy production accounted for about one-third of global GHG emissions in 2019 (IPCC, 2022), the household margin is macro-relevant. Yet engineering gains do not automatically translate into aggregate savings: adoption decisions are lumpy, prices and utilization respond endogenously, and informational or behavioral frictions can slow diffusion. As a result, higher average efficiency need not imply proportionate declines in energy use.

This paper asks: how responsive is household demand for energy-efficient durables to improvements in efficiency, and through which mechanisms? I develop a partial-equilibrium Bewley–Aiyagari model with incomplete markets, idiosyncratic income risk, fixed adjustment costs, and a discrete durable choice between a baseline and a more efficient type. Both durables produce services that require energy each period, but at different intensities; households choose utilization optimally. The efficient type commands a price premium along an empirically disciplined efficiency–price locus. Calibrated to the U.S. room air-conditioner market, the model matches ownership and expenditure moments and yields macro-relevant counterfactuals.

The main finding is that a 10% improvement in average efficiency raises adoption of the efficient type by 9% even when purchase prices rise accordingly; residential energy use falls by roughly 2%, and welfare increases. A decomposition attributes the aggregate response to three forces: (i) a utilization (rebound) channel—lower operating costs raise optimal services and the value of efficiency; (ii) selection—high-use households endogenously sort into efficient ownership; and (iii) inertia—fixed costs concentrate switching at replacement times.

Although the calibration relies on air-conditioner data, the mechanisms apply to a broad class

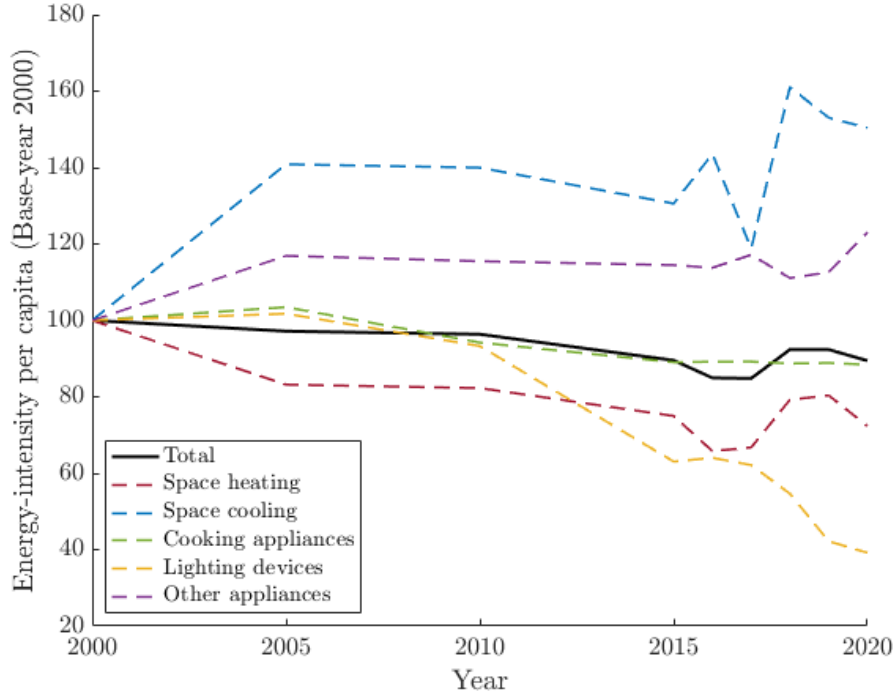
of household durables—HVAC, major appliances, and lighting—where service flows require energy and replacement is episodic.¹ I deliberately abstract from vehicle purchases, which have been analyzed extensively (e.g., Allcott (2011), Allcott, Mullainathan, and Taubinsky (2014), and Gavazza and Lanteri (2021)). Air-conditioners are an informative case: adoption is widespread but not universal—about 80% of U.S. households owned an air-conditioning device in 2020 (RECS; U.S. Energy Information Administration (2020))—lifetime costs are salient but typically not prohibitive, and market data are recent and detailed.

HVAC, large appliances, and lighting account for roughly 70% of residential energy consumption (U.S. Energy Information Administration (2015), henceforth RECS (2015)). Engineering measures of efficiency have generally improved since 2000, yet per-capita residential energy intensity has not uniformly fallen over the same period (Figure 1). Forward-looking scenarios (IEA, 2022) place a sizable share of decarbonization on continued efficiency progress; Figure 2 contrasts observed air-conditioner efficiency with the Sustainable Development Scenario (SDS). Taken together, the figures underscore that efficiency improvements alone are insufficient to guarantee lower aggregate intensity once utilization, composition, and adoption frictions are considered.

This paper is related to two strands of the literature. First, it contributes to the literature on energy-efficiency choices, which seeks to understand what motivates households’ efficiency decisions (e.g. Hausman (1979), Anderson and Claxton (1982), Dubin and McFadden (1984), Ruderman, Levine, and McMahon (1987), Gaspar and Antunes (2011), Mills and Schleich (2012), Gromet, Kunreuther, and Larrick (2013), Sallee (2014), Newell and Siikamäki (2015), Ramos et al. (2015)) and investigates the existence of an energy-efficiency gap, i.e. the hypothesis that households’ efficiency investments do not usually minimize their lifetime cost (e.g. Hirst and Brown (1990), Jaffe and Stavins (1994), Gerarden, Newell, and Stavins (2017)). While this literature typically relies on surveys and reduced-form analyses, my results stem from a structural model of discrete choice between energy-using durables with

¹HVAC stands for Heating, Ventilation and Air Conditioning.

Figure 1: US Per-capita Energy-intensity by end-use

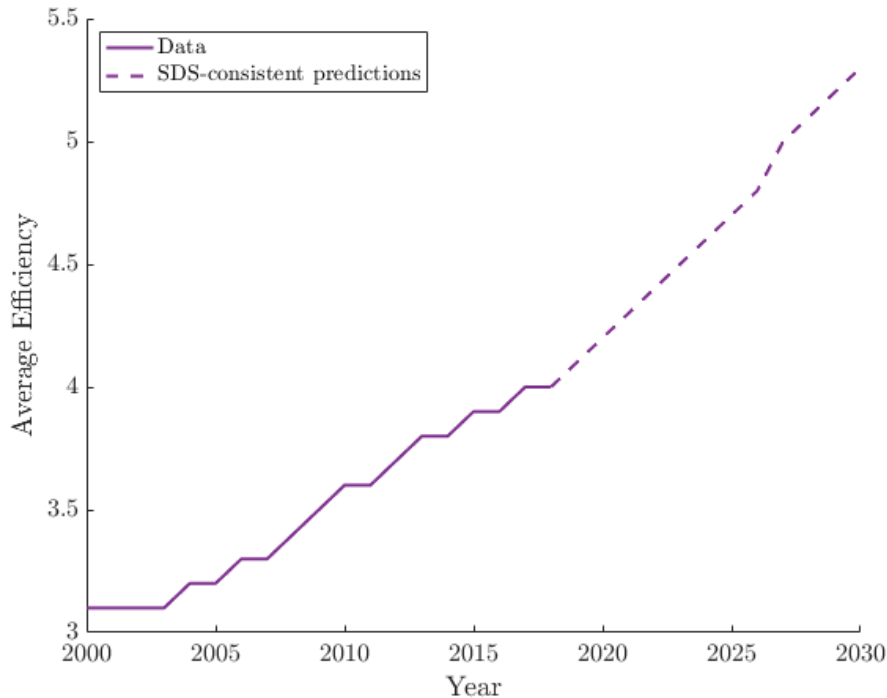


Note: This figure shows the evolution of the per-capita energy-intensity of the residential sector, in the US, between 2000 and 2020, taking 2000 as a base-year. Per-capita energy-intensity is the ratio of the total amount of energy used (here, in GJ) by the residential sector, and the population size at that time. The black line shows the overall energy-intensity, while the coloured lines distinguish between various end-uses. (Source: IEA Energy Efficiency Indicators, 2022)

a fixed adjustment cost, designed to capture the key frictions described in the literature, in an agnostic manner.

Four non-pecuniary forces have been found to influence efficiency decisions. First, inertia: merely owning a good significantly reduces the likelihood to invest in a new one, even if doing so would reduce total expenditure (Anderson and Claxton, 1982). Second, imperfect information about energy efficiency or energy prices are believed to prevent households from systematically making the optimal investment decision (Anderson and Claxton, 1982, Ramos et al. (2015)). Third, home-ownership: landlords have been found to be about 40% more likely to invest in an efficient durable than renters (Gaspar and Antunes, 2011). Fourth, behavioural biases, namely rational inattention (Sallee, 2014) and myopia. Unexpectedly, more patient agents exhibit a higher willingness-to-pay for efficient durables (Newell and

Figure 2: Observed and SDS-consistent global air-conditioner average efficiency



Note: This figure shows the evolution of the average energy efficiency of air-conditioners in use (solid line), and, from 2019, the evolution of the average efficiency that is consistent with the IEA’s Sustainable Development Scenario (dashed line). The efficiency is measured by the average CEER (combined Energy-efficiency ratio) of air-conditioners currently owned by the residential sector. A CEER is the ratio of the cooling output (usually in BTU/h) to the input of energy (usually in watts/h). A higher ratio implies greater efficiency. (Source: IEA Energy Efficiency Indicators, 2022)

Siikamäki, 2015).

Second, this paper is connected to the macroeconomic literature on lumpy durable investments (e.g. Berger and Vavra (2015), Gavazza and Lanteri (2021), Winberry (2021) or McKay and Wieland (2021)). In particular, it is most related to Lanteri and Rampini (2023), which studies firms’ efficiency decisions in a dynamic setting, and finds that larger firms are more likely to invest in efficient capital. I focus here on households’ investment.

The remainder is organized as follows. Section 2 documents the air-conditioner data from the U.S. Department of Energy. Section 3 presents the model. Section 4 details calibration and identification. Section 5 reports the elasticity and counterfactuals. Section 6 concludes.

2 Data

This section documents the empirical inputs used to discipline the model. The primary source is the U.S. Department of Energy’s Technical Support Document (TSD; Department of Energy (2023), henceforth DOE (2023)), which provides a comprehensive snapshot of the 2021 room air-conditioner (RAC) market. The TSD reports shipment shares by product class (PC), efficiency measures (CEER), average purchase prices, and expected lifetimes, allowing me to characterize both the price–efficiency trade-off and the distribution of technologies in use.

Product classification and coverage. DOE (2023) distinguishes sixteen product classes based on design (e.g., presence of a reverse cycle; louvered vs. non-louvered sides) and capacity (BTU/h). Because PCs 1–5 account for roughly 90% of total shipments, I focus on these classes in the baseline statistics and model targets. These PCs correspond to non-reverse cycle, louvered units spanning capacities from below 6,000 BTU/h to above 20,000 BTU/h. For completeness, Appendix A reports characteristics and market shares for all sixteen PCs.

Efficiency levels and aggregation. Within each PC, DOE reports market shares for six efficiency levels (EL0–EL5) aligned with the Energy Star program (initiated in 1992). EL0 and EL1 correspond to non-Energy Star units; EL2–EL5 correspond to Energy Star units, from least to most efficient.² I aggregate across PCs to construct shipment-weighted moments by efficiency level: CEER, price, and annual electricity use (kWh). Annual kWh figures assume 785 hours of operation per year, following DOE’s convention. Table 1 reports the resulting aggregates; Appendix B provides the underlying PC-by-EL data.

Moments for calibration and validation. For the structural analysis, I collapse ELs into two economically meaningful types: baseline (EL0–EL1) and Energy Star (EL2–EL5).

²Energy performance is measured by CEER (combined energy–efficiency ratio), the ratio of cooling output (BTU/h) to electrical input (W/h); higher CEER indicates greater efficiency.

Table 1: DOE data summary

Efficiency level	Data					
	0	1	2	3	4	5
Shipment-weighted CEER	10.9	11.3	12.0	13.7	15.9	20.9
Shipment-weighted price	506.5	508.8	516.3	526.5	612.3	654.3
Shipment-weighted kWh	596.1	576.5	521.6	456.6	389.2	296.6
Market share	0.03	0.56	0.32	0.01	0.08	0.00

Note: Shipment-weighted characteristics by efficiency level (EL). EL0–EL1 are non–Energy Star; EL2–EL5 are Energy Star. Prices are 2021 USD. Annual kWh assume 785 hours of operation. Source: DOE (2023).

Table 2 reports the resulting shipment-weighted moments. Relative to baseline, Energy Star units are about 5% more expensive and consume roughly 14% less electricity on average; baseline models represent about 59% of shipments. These moments discipline the relative price premium, the operating-cost contrast, and the cross-sectional composition targeted in calibration.

Table 2: Baseline and Energy Star air-conditioners

Variable	Type	
	Baseline	Energy Star
Shipment-weighted CEER	11.30	12.74
Shipment-weighted price	508.67	534.30
Shipment-weighted kWh	577.54	495.82
Market share	0.59	0.41

Note: Aggregation of EL0–EL1 (baseline) and EL2–EL5 (Energy Star). Prices are 2021 USD. Source: DOE (2023).

3 Calibration

This section details how I discipline the parameters of the model and documents the quantitative implications of these choices. Four parameters are estimated directly from data ($\{p_1, \bar{\chi}, e_\infty, \alpha\}$) while the remaining objects are either calibrated from the literature or normalized. Throughout, a model period is one year and, when relevant, I align levels to

2021 values to match the DOE cross-section used in Section ?? . The guiding principle is to anchor (i) intertemporal choices and idiosyncratic risk to well-accepted estimates from the durable-investment literature and (ii) the technology–price mapping for air conditioners to the engineering-based efficiency–price schedule. The combination identifies the *user-cost* channel (qe_d) and the *lumpy adjustment* channel ($\bar{\chi}$) that jointly govern adoption and utilization.

Table 3: Partial equilibrium model parametrization

Symbol	Parameter	Value	Source
<i>Calibrated</i>			
β	Yearly discount rate	0.96	Berger and Vavra (2015)
r	Interest rate	0.05	Berger and Vavra (2015)
p_2	Cost – Efficient durable	$1.05 * p_1$	DOE (2023)
e_2	Energy intensity – Efficient	$0.86 * e_1$	DOE (2023)
ξ_1	Failure rate – Baseline durable	0.11	DOE (2023)
ξ_2	Failure rate – Efficient durable	0.11	DOE (2023)
ρ	Income process persistence	0.914	Berger and Vavra (2015)
σ	Income process std	0.216	Berger and Vavra (2015)
δ	Depreciation rate	0.06	Berger and Vavra (2015)
θ	Elasticity of substitution	0.5	McKay and Wieland (2021)
<i>Estimated</i>			
p_1	Cost – Baseline durable	0.1	GMM estimation
$\bar{\chi}$	Fixed cost upper bound	13.95	GMM estimation
e_∞	Energy intensity – Infinity cost	0.34	OLS regression
α	Efficiency–price locus curvature	0.21	OLS regression
<i>Assigned</i>			
w	Wage rate	0.5	—
q	Energy rate	0.5	—
p_0	Cost – No durable	0	—
e_0	Energy intensity – No durable	0	—
e_1	Energy intensity – Baseline durable	1	—
ξ_0	Failure rate – No durable	1	—
ϕ	Borrowing constraint	0	—

Note: Parameters are either taken from the literature, estimated from DOE (2023), or normalized to fix units. One model period is one year; where applicable, 2021 levels are used for consistency with the shipment-weighted DOE snapshots.

Parametrization. Table 3 summarizes the baseline values. I set the intertemporal block

$\{\beta, r, \rho, \sigma, \delta\}$ to standard estimates used in work on lumpy household durables (see Berger and Vavra (2015)). These choices pin down the precautionary-savings motive and replacement incentives independent of energy considerations. The elasticity $\theta = 0.5$ follows McKay and Wieland (2021) and delivers non-trivial, but not extreme, substitution between services and the non-energy composite; this matters because the rebound of utilization λ to a lower user cost qe_d scales with θ .

On the technology side, I discipline $\{p_2, e_2, \xi_1, \xi_2\}$ directly from DOE (2023). First, Energy Star units are, on average, about 5% more expensive than non-certified units in shipment-weighted prices, which motivates $p_2 = 1.05 p_1$. Second, average energy intensity for efficient units is about 14% lower than baseline, which I encode as $e_2 = 0.86 e_1$. Third, DOE reports similar expected lifetimes across efficiency levels (roughly 9.3 years); I therefore set $\xi_1 = \xi_2 = 0.11$. These choices ensure that differential adoption arises primarily from operating-cost savings versus up-front premia, not from mechanically different failure risks.

A few normalizations fix units. I take the composite good as the numeraire and normalize $e_1 = 1$; consequently, q scales the user cost in model units. Setting $p_0 = e_0 = 0$ keeps the option of “not purchasing this period” innocuous; setting $\xi_0 = 1$ ensures updaters that skip investment do not artificially accumulate in a no-durable absorbing state. Finally, I impose $\phi = 0$ so that borrowing is not used to fund large upgrades; this restriction is conservative for diffusion because it strengthens the role of cash-on-hand and the fixed cost in shaping inaction regions.

Estimation. Two parameters, p_1 and $\bar{\chi}$, are chosen by GMM to match (i) a consumption-to-income ratio of 0.83 and (ii) a baseline-share of 0.59. The first moment ties the overall level of nondurable expenditure to income net of energy outlays, thereby disciplining the scale of the composite good and the tightness of household budgets in steady state (CEX 2021 and RECS 2020).³ The second moment maps directly to the cross-sectional mix of

³From the Bureau of Labor Statistics 2021 Consumer Expenditure Survey, average post-tax income is \$78,743 and total average expenditure is \$66,928; RECS (2020) records average energy expenditure \$1,884,

baseline versus efficient units in DOE shipments and therefore identifies the interaction of the price premium and the fixed cost with heterogeneity in intended utilization. Intuitively, p_1 scales the price level of the durable, affecting the financial margin and the propensity to purchase at failure, while $\bar{\chi}$ governs the width of the inaction region and hence the share that voluntarily upgrades mid-life. Estimating these two parameters jointly makes the model replicate the aggregate penetration of efficient units without leaning on non-credible micro targets.

The resulting estimates are $p_1 = 0.1$ and $\bar{\chi} = 13.95$. In the stationary equilibrium these values deliver a consumption-to-income ratio of 0.86 and a baseline share of 0.5953, close to targets (Table 4). The small discrepancy in the first moment is consistent with the model’s precautionary saving and the fact that energy outlays are endogenous via λ .

Table 4: Targeted moments

Moment	Target	Model
Consumption-to-income ratio	0.83	0.86
Share of baseline durables	0.59	0.5953

Note: GMM chooses $(p_1, \bar{\chi})$ to match aggregate expenditure and the baseline share in DOE shipments. The first moment anchors the budget environment; the second fixes the extensive margin of efficiency adoption.

The remaining estimated parameters, e_∞ and α , govern the *efficiency-price locus* that maps engineering improvements into purchase premia. I estimate them from DOE (2023) by adapting the classic approach in Ruderman, Levine, and McMahon (1987). Writing the locus as

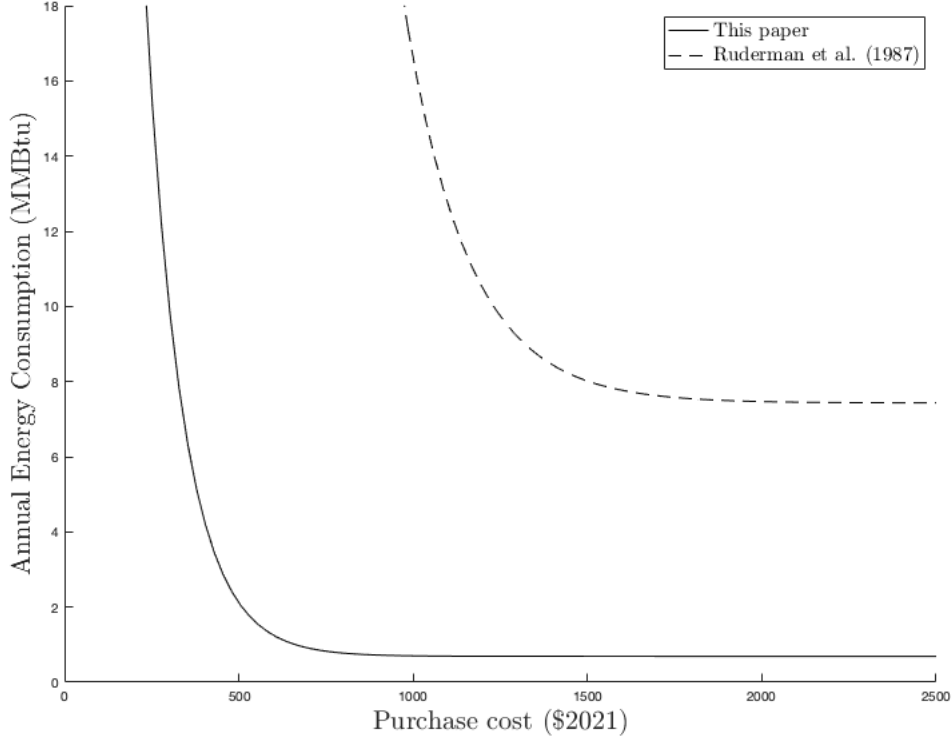
$$\frac{p_d}{p_1} - 1 = -\alpha \ln\left(\frac{e_d - e_\infty}{e_1 - e_\infty}\right) + \varepsilon_d,$$

I perform a grid search over e_∞ and, for each candidate, estimate α by OLS; I then select the pair that maximizes R^2 (see Corum and O’Neal (1982) for related procedures). The preferred estimates are $\alpha = 0.21$ (s.e. 0.02, $p < 0.001$) and $e_\infty = 0.69$, with $R^2 = 0.95$. Figure 3 plots

implying $(66,928 - 1,884)/78,743 \approx 0.83$.

the implied empirical locus together with the original Ruderman et al. schedule; the modern curve lies down and to the left, consistent with efficiency improving and premia compressing relative to the 1980s.⁴

Figure 3: Empirical efficiency–price locus

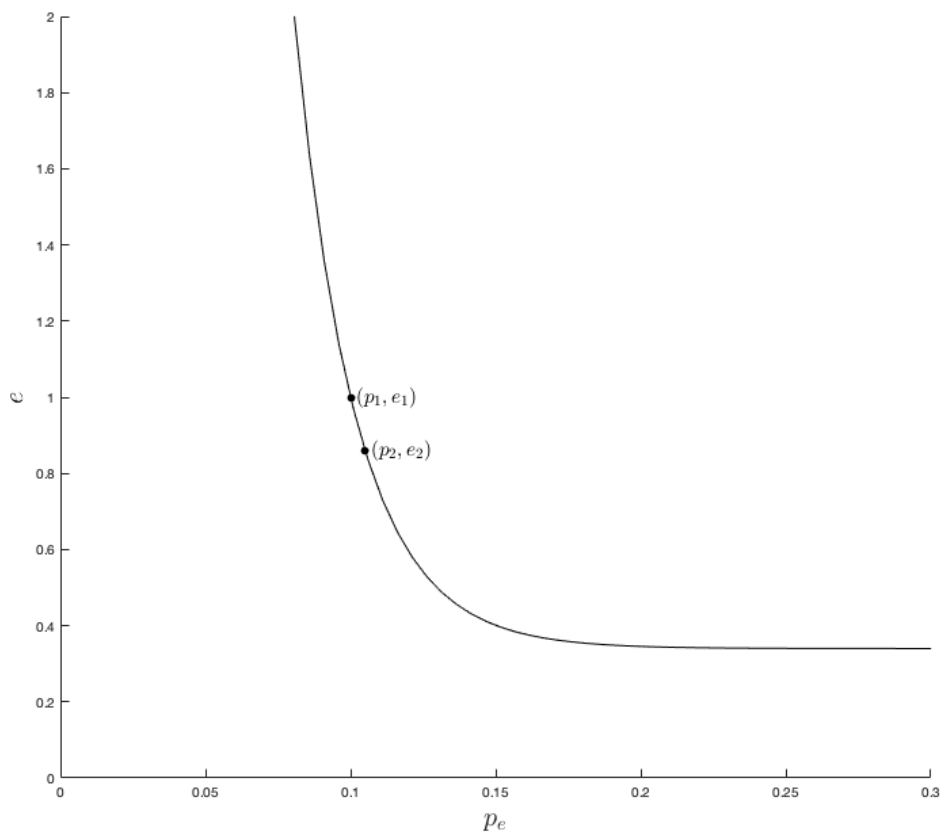


Note: Solid: DOE (2023) estimates; dashed: Ruderman, Levine, and McMahon (1987). Horizontal axis: purchase cost (2021 USD). Vertical axis: implied annual energy use (million BTUs).

For counterfactuals, I normalize the model’s baseline intensity to $e_1 = 1$ and select (α, e_∞) to preserve the empirical ratio e_∞/e_1 . This yields $\alpha = 0.21$ and $e_\infty = 0.34$ in model units. Figure 4 shows the resulting schedule together with the positions of (p_1, e_1) and (p_2, e_2) . Table 5 reports side-by-side the Ruderman et al. parameters, the DOE-based empirical estimates, and the model normalization. This mapping ensures that when I perturb e_2 in the policy experiments below, the associated price premium adjusts coherently rather than being imposed arbitrarily.

⁴This shift is consistent with diffusion and learning-by-doing in components, and with standards pushing the mass of products toward higher CEER.

Figure 4: Model efficiency–price locus



Note: The locus translates engineering intensity e into a purchase price premium relative to p_1 . Markers show the calibrated baseline and efficient device.

Policy functions in partial equilibrium. Figure 5 displays updater policy functions across assets for different income states. The durable choice panel exhibits a rich, non-monotone threshold structure: at median income, the efficient type is optimal across almost all wealth levels because intended utilization is high enough that energy savings dominate the premium; at low income, the efficient type is chosen only once assets are sufficient to support higher c (and thus λ), which raises the payoff to lower e ; at high income, households initially tolerate baseline operation because the marginal utility of saving on energy is low relative to the premium, but they switch as assets (and thereby utilization) increase further. These patterns are the mechanical consequence of the static condition $\lambda = c(qe_d)^{-\theta}$ interacting with the fixed cost: the private return to efficiency scales with use , not solely with income or

Table 5: Efficiency–price locus: empirical vs. model parameters

	e_1	e_∞	$\frac{e_\infty}{e_1}$	α
Ruderman et al.	15.69	7.43	0.47	0.18
DOE (2023)	2.03	0.69	0.34	0.21
Model	1	0.34	0.34	0.21

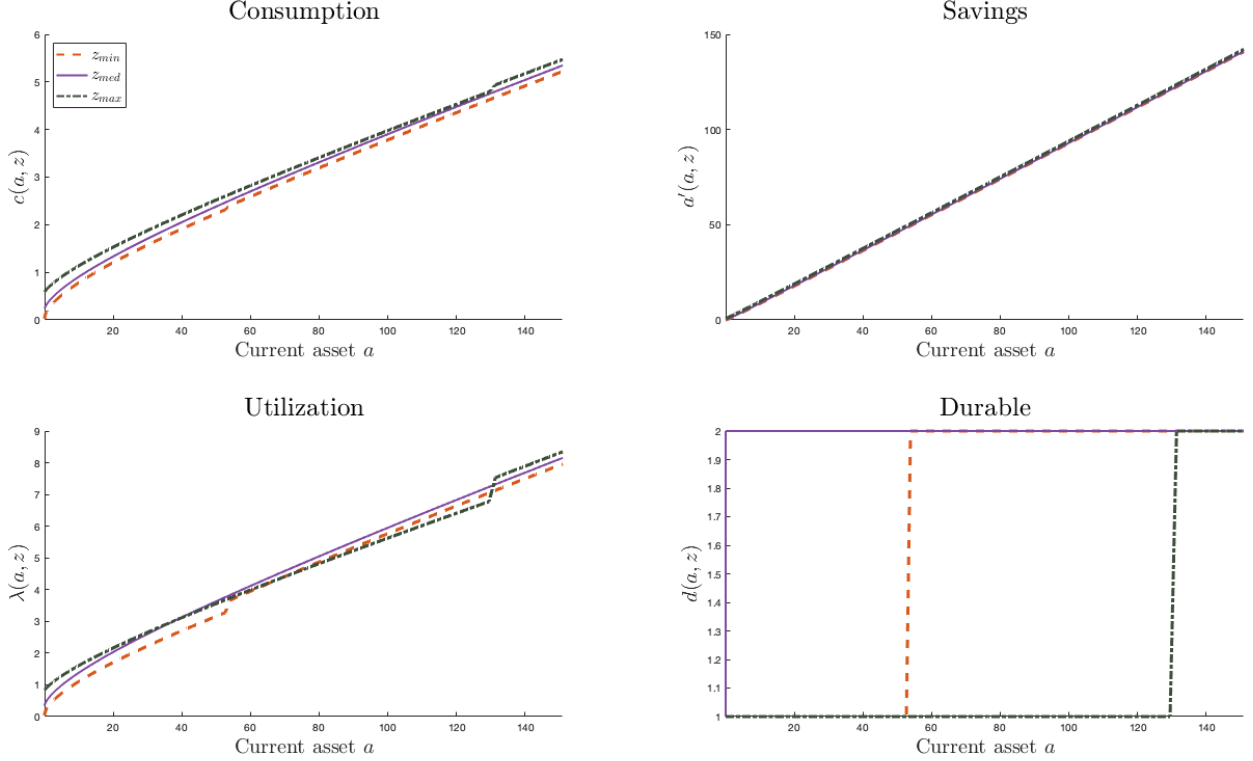
Note: The model preserves the empirical e_∞/e_1 ratio while normalizing $e_1 = 1$, so shifts in e_2 imply internally consistent premia.

wealth, and the option value of waiting smooths adjustments except when shocks are large or failures occur.

Figure 6 shows the corresponding policies for non-updaters by incumbent type. Conditional on keeping the device, higher-ability households choose higher c and, via the user-cost condition, higher λ . The gap in operating cost between e_1 and e_2 therefore widens in utility terms with (a, z) , which is why voluntary upgrading is concentrated in states with high intended use and why, in the stationary distribution, efficient devices are overrepresented among high-wealth households.

Stationary distributions in partial equilibrium. The left panel of Figure 7 reports the population split between updaters (13.01%) and non-updaters, closely mirroring RECS (2020) evidence that roughly 13% of households purchase air-conditioning equipment within a two-year window. The center and right panels display the wealth distribution by updater status and by incumbent type. Baseline devices are disproportionately held by poorer households, while efficient devices tilt toward higher-wealth bins, exactly as the selection-on-use mechanism implies. Because failures arrive at rate ξ_d regardless of type, the stock composition in steady state is not mechanically driven by differential lifetimes; rather, it emerges from the interaction of the price premium, operating savings, and the endogenous inaction region.

Figure 5: Policy functions – Updaters



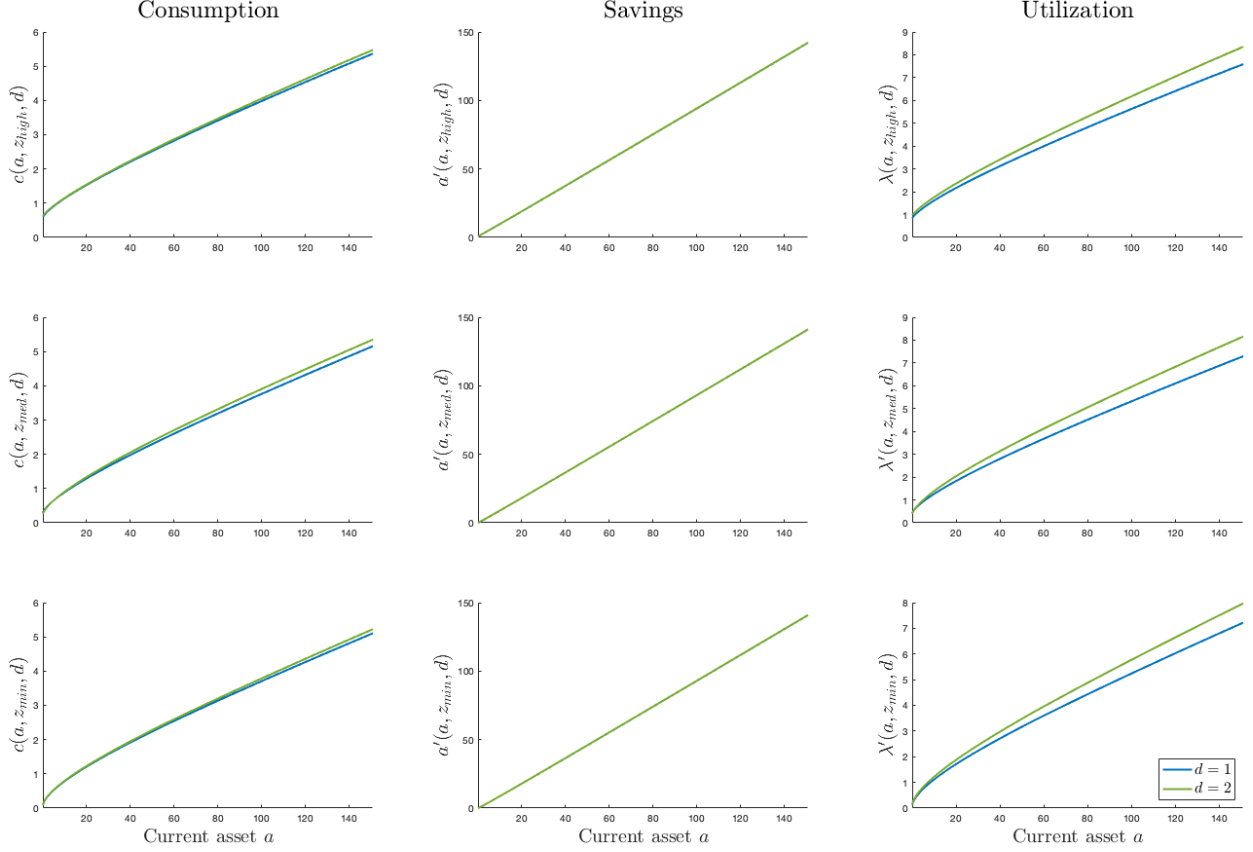
Note: Each curve shows the optimal policy as a function of assets a for a given ability z . The fourth panel reports the discrete durable choice.

4 The Energy-efficiency elasticity of demand

Experiment. To estimate the energy-efficiency elasticity of demand, I conduct two comparative statics experiments. Starting from the calibrated stationary partial equilibrium, I first decrease the energy intensity of the efficient good, e_2 , by 10%, keeping everything else constant. Doing so is akin to increasing the energy-efficiency of the efficient good, since it now consumes less energy per unit of time. In a second time, I decrease the energy intensity of the efficient good e_2 and accordingly change the price of the efficient good, using the estimated efficiency-price locus estimated above.

Results. The main results are summarized in Table 6. The first column corresponds to the initial partial equilibrium; the second column corresponds to the first experiment, which

Figure 6: Policy functions – Non-updaters



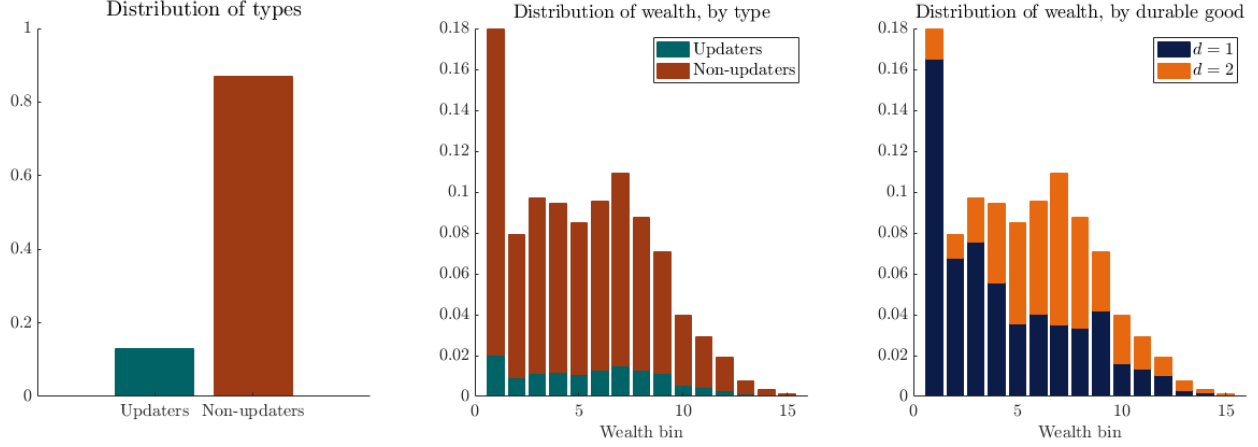
Note: Rows correspond to high/median/low ability z ; each panel traces the policy over assets for each incumbent durable type d .

consisted in only decreasing e_2 ; the last column corresponds to the second experiment, which consisted in decreasing e_2 and adjusting p_2 accordingly, given my estimation of the efficiency-price locus above.

Both experiments yield similar results. In particular, a 10% decrease in e_2 decreases the share of baseline durables by 9%, from 60% to 54% even if the price of the efficient good increases. Energy use decreases by about 2% in both cases. Consumption increases, since the more efficient durable allows household to substitute utilization for consumption. Overall, welfare increases while energy consumption decreases.

Given the model, I therefore estimate that a 10% increase in the average energy efficiency (or, similarly, a 10% decrease in the average energy intensity) would increase households' invest-

Figure 7: Partial stationary equilibrium distributions



Note: Left: share of updaters vs. non-updaters. Middle: wealth distribution by updater status. Right: wealth distribution by incumbent durable type.

ment in efficient durables by about 9%. Therefore, households' energy-efficiency elasticity for efficient goods is high.

Table 6: Comparative statics: Results

Variable	(1)		(2)		(2')	
	e_2	p_2	e_2	p_2	e_2	p_2
	0.86	0.1050	0.7740	0.1050	0.7740	0.1088
Energy	0.4344		0.4265		0.4262	
Consumption	0.3163		0.3190		0.3188	
Share of baseline durables	0.5953		0.5417		0.5418	
Welfare	-38.3991		-37.9951		-38.0099	

Note: This table summarizes the results of the comparative statics experiment. The first column shows aggregate values obtained by the model in the initial stationary partial equilibrium. Column (2) shows how these aggregate values change if e_2 , i.e. the energy intensity of the efficient durable good, decreases by 10%. Column (3) shows how the aggregate values change if the price of the efficient durable good, p_2 , adjusts to the lower e_2 .

5 Conclusion

Despite the potential benefits associated with greater energy-efficiency, the way households make efficiency decisions is little understood. Therefore, it is difficult to devise policies to

properly incentive greater energy-efficiency for households.

This paper proposed a structural model capturing the main trade-offs faced by households when making an efficiency investment decision. Using data on the air-conditioner market, I estimate that households' efficiency-elasticity for efficient energy-using durables is high: a 10% increase in energy efficiency leads to a 9% increase in the share of efficient durables. In addition, greater efficiency leads to greater welfare, and lower energy use.

The analysis described above all relied on a partial equilibrium setup. A natural next step is to embed the model used here in a general equilibrium setting, thereby accounting for potential general equilibrium effects. At least two such effects may be important. First, a decrease in energy consumption might reverberate onto the energy market, thus affecting energy prices. This effect would be particularly salient if the change in households' demand for energy decreased significantly as a share of total energy demand. Second, while energy-using durable prices have been taken as given here, households may foresee a future increase in efficiency, and as such an associated increase in prices, and adjust their current decisions accordingly. This phenomenon is not captured in this model but should be further examined.

Overall, the result of this paper illustrates how auspicious energy-efficiency is, since it could allow for a significant decrease in energy-production, with at worst no effect, and at best a positive impact on welfare.

6 Appendix

A. Room air-conditioner Product Classes (PCs)

PC	Characteristics	BTU/h	Market share
1	Without reverse cycle, with louvered sides	< 6,000	89.7%
2	Without reverse cycle, with louvered sides	6,000 – 7,990	
3	Without reverse cycle, with louvered sides	8,000 – 13,990	
4	Without reverse cycle, with louvered sides	14,000 – 19,990	
5	Without reverse cycle, with louvered sides	> 20,000	
6	Without reverse cycle, without louvered sides	< 6,000	8.7%
7	Without reverse cycle, without louvered sides	6,000 – 7,990	
8	Without reverse cycle, without louvered sides	8,000 – 13,990	
9	Without reverse cycle, without louvered sides	14,000 – 19,990	
10	Without reverse cycle, without louvered sides	> 20,000	
11	With reverse cycle, with louvered sides	< 20,000	0.9%
12	With reverse cycle, with louvered sides	> 20,000	
13	With reverse cycle, without louvered sides	< 14,000	0.3%
14	With reverse cycle, without louvered sides	> 14,000	
15	Casement	Casement only	0.4%
16	Casement	Casement slider	

Source: DOE (2023)

B. Room air-conditioner characteristics, by Product Class (PC) and Efficiency Level (EL)

PC	Efficiency Level	CEER	Cost (\$)	Average lifetime	EL Market share (within PC)
1	0	11	419.43	9.3	0.077
1	1	11.4	421.09	9.3	0.852
1	2	12.1	424.44	9.3	0.021
1	3	13.1	429.24	9.3	0
1	4	16	517.69	9.3	0.05
1	5	20.2	532.09	9.3	0
2	0	11	437.31	9.3	0
2	1	11.4	439.94	9.3	0.746
2	2	12.1	444.07	9.3	0.183
2	3	13.7	463.46	9.3	0.021
2	4	16	538.89	9.3	0.05
2	5	21.2	599.43	9.3	0
3	0	10.9	560.99	9.3	0
3	1	11.4	563.63	9.3	0.303
3	2	12	576.41	9.3	0.58
3	3	14.3	584.35	9.3	0.009
3	4	16	668.76	9.3	0.107
3	5	21.9	727.46	9.3	0
4	0	10.7	702.58	9.3	0
4	1	11.1	704.88	9.3	0
4	2	11.8	713.24	9.3	0.891
4	3	14	738.67	9.3	0.001
4	4	16	834.56	9.3	0.107
4	5	19.8	867.85	9.3	0
5a	0	9.4	875.77	9.3	0
5a	1	9.8	878.92	9.3	0.09
5a	2	10.3	893.1	9.3	0.803
5a	3	11.8	908.96	9.3	0
5a	4	13.8	1014.33	9.3	0.107
5a	5	18.7	1056.5	9.3	0
5b	0	9	926.16	9.3	0.403
5b	1	9.4	929.42	9.3	0.457
5b	2	9.9	935.12	9.3	0.09
5b	3	10.3	939.12	9.3	0
5b	4	13.2	1079.52	9.3	0.05
5b	5	16.3	1105.98	9.3	0

Source: DOE (2023)

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